

7.4

$p\text{-value}=0.02$	(c)	(a) do not reject H_0 at $\alpha = 0.1$
$p\text{-value}=0.07$	(b)	(b) reject H_0 at $\alpha = 0.1$ but not at 0.05
$p\text{-value}= 0.3$	(a)	(a) reject H_0 at $\alpha = 0.05$ but not at 0.01
$p\text{-value}= 0.006$	(d)	(d) reject H_0 at $\alpha = 0.01$

7.11

- (a) $H_0: p \leq 0.3$ versus $H_1: p > 0.3$
- (b) When $p = 0.25$, H_0 is true. Let X be the number of patients who survived after being treated by the new drug. Then $X \sim \text{Bin}(15, 0.25)$ and the type I error probability is

$$P(X \geq 8 \text{ given } p = 0.25) = 1 - P(X \leq 7) = 1 - 0.9827 = 0.0173.$$
- (c) When $p = 0.3$, H_0 is true. Let X be the number of patients who survived after treated by the new drug. Then $X \sim \text{Bin}(15, 0.3)$ and the type I error probability is

$$P(X \geq 8 \text{ given } p = 0.3) = 1 - P(X \leq 7) = 1 - 0.95 = 0.05.$$
- (d) When $p = 0.4$, H_1 is true. Let X be the number of patients who survived after treated by the new drug. Then $X \sim \text{Bin}(15, 0.4)$ and the type II error probability is

$$P(X < 8 \text{ given } p = 0.4) = P(X \leq 7) = 0.7869.$$
- (e) From part (c), the significance level is 0.05.

7.13

- (a) $H_0: \mu \leq 200,000$ versus $H_1: \mu > 200,000$
- (b) Let X be the lifetime of an engine. Then under H_0 ,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N\left(200,000, \frac{30,000^2}{16}\right) = N(200,000, 7,500^2).$$

The type I error probability is

$$\begin{aligned} \alpha &= P(\text{reject } H_0 \text{ when } \mu = 200,000) = P(\bar{X} \geq 215,000) \\ &= P\left(Z \geq \frac{215,000 - 200,000}{7500}\right) = P(Z > 2) = 1 - \Phi(2) = 1 - 0.9772 \\ &= 0.0228. \end{aligned}$$

- (c)

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) = N(210,000, 7,500^2).$$

The type II error probability is

$$\begin{aligned} \beta(210,000) &= P(\text{do not reject } H_0 \text{ when } \mu = 210,000) = P(\bar{X} < 215,000) \\ &= P\left(Z < \frac{215,000 - 210,000}{7500}\right) = \Phi(0.6667) = 0.7475. \end{aligned}$$

8.5

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 = \left(\frac{z_{0.05} \sigma}{E} \right)^2 = \left(\frac{1.645 \cdot 15}{3} \right)^2 = 67.7$$

The required sample size is $n = 68$.

8.11

(a)

$$\begin{aligned} \beta(62) &= \Phi\left(z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) - \Phi\left(-z_{\alpha/2} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) \\ &= \Phi\left(2.58 + \frac{60 - 62}{4/\sqrt{25}}\right) - \Phi\left(-2.58 + \frac{60 - 62}{4/\sqrt{25}}\right) = \Phi(0.08) - \Phi(-5.08) \\ &= 0.5319 - 0 = 0.5319 \end{aligned}$$

(b)

$$n = \left(\frac{\sigma(z_{\alpha/2} + z_{\beta})}{\mu_0 - \mu'} \right)^2 = \left(\frac{\sigma(z_{0.005} + z_{0.1})}{\mu_0 - \mu'} \right)^2 = \left(\frac{4(2.58 + 1.28)}{60 - 62} \right)^2 = 59.6$$

The required sample size is 60.

8.14

(a) $H_0: \mu = 14$ versus $H_1: \mu < 14$

$n = 60$ is large.

The test statistic is

$$Z = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $z < -z_{\alpha} = -z_{0.05} = -1.645$.

(b)

$$z = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = \frac{13.5 - 14}{\frac{3}{\sqrt{60}}} = -1.29 > -1.645$$

Do not reject H_0 .

(c)

$$p\text{-value} = P(Z < -1.29) = 0.0985 > 0.05$$

Do not reject H_0 .

(d)

$$\begin{aligned} \beta(13) &= 1 - \Phi\left(-z_{\alpha} + \frac{\mu_0 - \mu'}{\sigma/\sqrt{n}}\right) = 1 - \Phi\left(-1.645 + \frac{14 - 13}{3/\sqrt{60}}\right) \\ &= 1 - \Phi(0.937) = 1 - 0.8256 = 0.1744 \end{aligned}$$

8.17

$n = 60$ is large. $\bar{x} = 70,300$, $s = 6,900$

(a)

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}} = 70,300 \pm 1.96 \frac{6,900}{\sqrt{60}} = 70,300 \pm 1,746$$

A 95% confidence interval is (68,554, 72,046).

(b)

$$\bar{x} \pm z_{\alpha/2} \frac{s}{\sqrt{n}} = 70,300 \pm 1.645 \frac{6,900}{\sqrt{60}} = 70,300 \pm 1,465$$

A 95% confidence interval is (68,835, 71,765).

8.22

$n = 8$ is small. $\bar{x} = 25.425$, $s = 2.542$

(a)

$$\bar{x} \pm t_{\frac{\alpha}{2}, n-1} \frac{s}{\sqrt{n}} = \bar{x} \pm t_{0.025, 7} \frac{s}{\sqrt{n}} = 25.425 \pm 2.365 \frac{2.542}{\sqrt{8}} = 25.425 \pm 2.126$$

A 95% confidence interval is (23.30, 27.55).

(b)

$$H_0: \mu \leq 25, H_1: \mu > 25$$

$$\alpha = 0.05$$

The test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $t \geq t_{\alpha, n-1} = t_{0.05, 7} = 1.895$.

$$t = \frac{25.425 - 25}{2.542/\sqrt{8}} = 0.473 < 1.895$$

Do not reject H_0 .

$$p\text{-value} = P(T > 0.473) = 0.325$$

(c) Type II error

8.25

$n = 13$ is small. $\bar{x} = 129.62$, $s = 1.66$

$$H_0: \mu = 130, H_1: \mu \neq 130$$

$$\alpha = 0.05$$

The test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $|t| \geq t_{\alpha/2, n-1} = t_{0.025, 12} = 2.179$.

$$t = \frac{129.62 - 130}{1.66/\sqrt{13}} = -0.8254$$

Since $|t| = 0.8254 < 2.179$, we do not reject H_0 .

$$p\text{-value} = P(|T| > 0.8254) = 2P(T > 0.8254) = 2(0.2126) = 0.4252$$

8.26

$n = 14$ is small. $\bar{x} = 61.18$, $s = 3.68$

(a)

$$H_0: \mu \geq 63, H_1: \mu < 63$$

$$\alpha = 0.01$$

The test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

The rejection region is $t \leq -t_{\alpha, n-1} = -t_{0.01, 13} = -2.650$.

$$t = \frac{61.18 - 63}{3.68/\sqrt{14}} = -1.85 > -2.650$$

Do not reject H_0 .

(b) $p\text{-value} = P(T < -1.85) = 0.0436$

(c) Type II error